Predicting Linear Relational Decodability Without the Relation

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Abstract

Linear relational embeddings (LREs) approximate a language model's subject-to-object mapping with an affine function. Their faithfulness varies enormously across relations – near zero for PERSON_FATHER, at ceiling for COUNTRY_CAPITAL_CITY. We ask what determines that variation and show it is substantially predictable from a readout that never sees the relation. We estimate a novel form of a Jacobian lens, the *differential Jacobian lens*: instead of model outputs for "France", we instead contrastively predict outputs for "France" that are not also outputs for "Spain". After estimating this differential Jacobian lens, we measure how highly it ranks the model's answer for a relation – without using any relation-specific information in estimation and prediction. Across 28 relations of the LRE dataset on Olmo-3-7B this relation-agnostic score correlates with LRE faithfulness at $\rho = 0.66$ ($p < .01$). This suggests that linear decodability may be largely a property of what a subject's representation already carries, rather than of the relation being decoded. After attempting to move the unit of analysis from the relation to the individual fact, we conclude that new relation datasets and analysis methods are needed to address several problems in existing datasets, most prominently semantically weak first tokens (e.g., most predicted universities' first token is "University") and subject ambiguity (e.g., the LLM associates obscure actor names with non-actor entities).

1 Introduction

Hernandez et al. (2024) showed that a language model's mapping from a subject to the object of a relation is often well approximated by an affine function of the subject's hidden state – a *linear relational embedding* (LRE). They also showed that this works far better for some relations than others. On Olmo-3-7B, we reproduce the same spread: an LRE fitted on COUNTRY_CAPITAL_CITY agrees with the model's own prediction 75% of the time while one fitted on PERSON_FATHER agrees 9% of the time.

Why? The natural reading is that some relations are "more linear" than others – that linearity is a property of the relation. We report evidence that it is better understood as a property of the *subject's representation*. Concretely, we show that one can predict how faithful an LRE will be *before fitting it*, and without using the relation at all.

Building on recent work by (Gurnee et al., 2026), we introduce the *differential Jacobian lens*, which evaluates the Jacobian at individual reference entities and applies it to *differences* between entity representations: *France* minus *Spain* rather than *France* alone. What the subtraction leaves is what is specific to this entity among others of its kind, and the Jacobian carries that residual forward to the vocabulary. The construction changes what the readout is about: from a category to an individual. The question it answers – *what is this model ready to say about this entity, as against others of its kind?* – is not specific to relations.

We estimate the differential Jacobian lens under a single neutral prompt shared by every relation, at one layer fixed in advance, and ask how highly it ranks the model's own answer. The lens sees no relation template, no objects and no (subject, object) pairs, yet its score correlates with LRE faithfulness across 28 relations at $\rho = 0.66$.

We make four contributions. (i) We introduce the differential Jacobian lens, a supervision-free readout of what a model is disposed to say about an individual entity. (ii) We turn it into a relation-agnostic diagnostic of linear decodability and show that it predicts LRE faithfulness. (iii) We show the prediction holds up under leave-one-domain-out and under controls for object cardinality and answer-token frequency. (iv) After attempting to move the unit of analysis from the relation to the individual fact, we provide an analysis of several problems in existing datasets, most prominently first-token

ambiguity of objects and reference prominence of subjects, that hinder further progress on understanding LREs in language models.

2 Background

An LRE for relation r approximates the model's computation as

$$o \approx \beta W s + b, \tag{1}$$

where s is the hidden state at the subject's final token and o the output state. Write f for the model's own map between the two: f sends the subject's hidden state at an intermediate layer ℓ to the final-layer state at the last position, holding the rest of the prompt fixed. The LRE is the first-order approximation of f, averaged over a small set $\{s_i\}$ of reference subjects,

$$W = \mathrm{mean}_i\, J_i, \qquad b = \mathrm{mean}_i[\, f(s_i) - J_i s_i \,], \tag{2}$$

where J_i is the Jacobian of f at s_i (Hernandez et al., 2024).

The scalar β corrects a systematic weakness of the first-order estimate. A Jacobian taken at a reference subject describes the model's map only in a neighborhood of that subject; extrapolated to a different subject it points in roughly the right direction but is too short. This matters because the prediction decomposes into a subject-dependent term and a constant: expanding b gives

$$\beta W s + b \;=\; \mathrm{mean}_i[\, f(s_i) + \beta J_i (s - s_i) \,], \tag{3}$$

in which only the second term depends on which subject is being decoded. Left unscaled, the shared anchor $\mathrm{mean}_i\, f(s_i)$ dominates and the operator returns much the same object for every subject. Setting $\beta > 1$ amplifies the subject-specific contribution relative to that anchor, and so does change which object is ranked first. Hernandez et al. (2024) fit β per relation alongside the layer; we follow their protocol and select it from $\{0.5, 1, 1.5, 2, 3, 5\}$. Eq. 3 is also what makes the LRE affordable: each test subject costs n Jacobian–vector products rather than a materialized 4096×4096 matrix.

Faithfulness is the rate at which decoding $\beta W s + b$ over the full vocabulary reproduces the *model's own* top-1 prediction – not the gold object. It is therefore a measure of how well an affine map captures what the model does. Faithfulness defined this way is the quantity we predict.

3 A relation-agnostic Jacobian lens

For an entity e (e.g., France) and a set of reference entities $\{i\}$ of the same type (e.g., Spain, Italy), we take the Jacobian–vector product at each reference and average:

$$v_e \;=\; \frac{1}{n} \sum_i J_i\, (s_e - s_i). \tag{4}$$

Here J_i is the Jacobian of the model's map from the subject's hidden state at layer ℓ to the final-layer state at the last position, evaluated at reference i. As discussed above, subtracting a same-type reference cancels it, leaving the entity-specific residual, and the Jacobian then carries that residual forward to the vocabulary. Our differential Jacobian lens therefore reads what this entity, as opposed to others of its kind, makes the model disposed to say.

We never form J_i explicitly, but obtain each Jacobian–vector product $J_i u$ by the standard double-backward construction (Pearlmutter, 1994), at the cost of a constant number of forward and backward passes; this is what makes Eq. 4 affordable at $d_{\text{model}} = 4096$ (Appendix A). Each J_i is evaluated under a *neutral* prompt, `"{} is known for"`, identical for every relation.

We decode v_e over the full vocabulary and record the rank of m_e, the model's own top-1 answer for relation r. Writing $\mathrm{rk}_{v_e}(m_e)$ for that rank, the diagnostic for a relation is

$$A_r \;=\; -\,\mathrm{median}_e\, \log_{10} \mathrm{rk}_{v_e}(m_e), \tag{5}$$

so higher A_r means the lens agrees with the model more.

Three properties make A_r non-trivial as a predictor. The prompt is neutral and shared, so no relation template enters. The differences in Eq. 4 are subject-minus-subject – *France* minus *Spain*, never *France* minus *Paris* – so no relation-specific (subject, object) pair is used in estimation. The layer is not relation-specific either. It is fixed at 17 for all relations (Appendix B).

Two prompts. Our experiments use two different prompts, for two different purposes. The *neutral prompt*, `"{} is known for"`, names no relation and is the same for all 28 relations; it is the only prompt the lens ever sees, and it is where the hidden states in Eq. 4 are taken. The *in-context prompt* is relation-specific: eight (subject, object) examples of the relation in question, followed by the query subject without its object, so that the model's next

token is its answer (§4). It is used for the LRE and for the model's own prediction, never for the lens.

Relation-specificity therefore enters in exactly one place: the *target*. What the lens is scored against is the model's top-1 under the in-context prompt – the same quantity faithfulness is defined against – so the two are commensurable, while everything on the lens side of the comparison remains relation-agnostic.

4 Experimental setup

Model. Olmo-3-7B, in `float32` with eager attention, as the Jacobian–vector products require both (Appendix A).

Relations. The LRE dataset (Hernandez et al., 2024) contains 47 relations. Our prompt presupposes an entity subject, so we fixed a subject-type filter in advance using the dataset's declared `domain_name` field: the 28 relations with named-entity subjects (*person*, *country*, *name*, *company*, *landmark*, *superhero*, *city*, *pokemon*, *product*, *star*). Excluded are relations over words as words (*verb*, *adjective*, *word*) and over abstracts (*occupation*, *task*, *characteristic*, *degree*), for which `"give is known for"` is a category error. No other relation was excluded.

Measurement. Both quantities are read off the same prompt, so that they are comparable. Following Hernandez et al. (2024), a relation is presented to the model *in context*: eight (subject, object) examples of that relation are concatenated, one per line, and the query subject is appended without its object, so that the model's next token is its answer. This is what defines "the model's own prediction" for both measures. The subjects of a relation are shuffled once and partitioned into three disjoint sets: eight in-context examples, eight reference subjects from which W and b are estimated, and the remainder as scored test subjects.

Faithfulness is then recomputed for all 28 relations under this protocol. Layer and β are chosen on a 6×6 grid – six candidate layers for the subject state $(5, 11, 14, 17, 20, 23)$ crossed with six values of β $(0.5, 1, 1.5, 2, 3, 5)$ – by the mean over the test subjects. Hernandez et al. (2024) likewise fit both per relation; the six layers span the network at roughly even spacing and the β range brackets the values they report.

The lens scores 11–20 subjects per relation, depending on how many remain once the in-context and reference sets are removed. For the results in §5, the lens and the LRE sample their splits independently, so the particular subjects entering each set differ between the two measurements. Both estimates are unbiased for the relation, but each carries sampling noise the other does not, which attenuates rather than inflates the correlation we report.

Crucially, the in-context prompt is used only to obtain the *target*; the lens itself is always computed under the neutral prompt of §3.

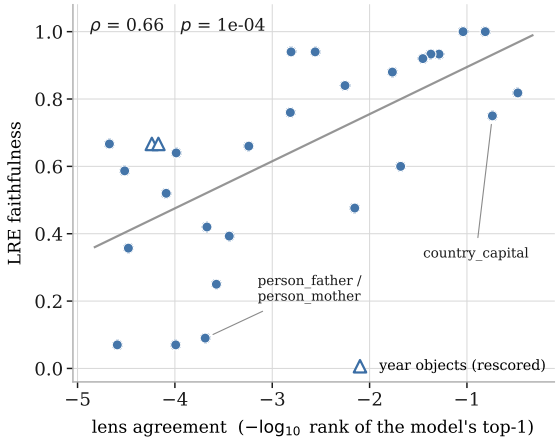


Figure 1: LRE faithfulness against the relation-agnostic lens score, 28 relations. The lens sees no relation template, no objects, and no (subject, object) pairs, and uses one fixed layer throughout. Triangles mark the two relations with numeric objects, rescored as described in Appendix ?.

5 Results

Figure 1 shows the result. The relation-agnostic score predicts LRE faithfulness at Spearman $\rho = 0.658$ ($p = 1.4 \times 10^{-4}$); a permutation test agrees (Appendix C).

The correlation is not carried by cardinality. Relations differ greatly in how many distinct objects they have; k, the mean number of subjects per object, ranges from 1.0 to 189. High k makes faithfulness easier, and indeed k predicts faithfulness on its own ($\rho = 0.43$, $p = 0.02$). But k is essentially uncorrelated with the lens score ($\rho = -0.03$). Removing k *raises* the correlation to $\rho = 0.744$ ($p = 10^{-5}$): k and the lens are independent predictors of faithfulness, so removing the variance attributable to k sharpens rather than weakens the

lens signal.[1]

Nor by answer frequency. A relation whose answers are frequent tokens might be easy for both measures. We estimate the model's prior over each answer token by averaging its log-probability across ten neutral contexts. This predicts neither faithfulness ($\rho = -0.03$) nor the lens score ($\rho = -0.02$), and removing it leaves the correlation unchanged at 0.658; The correlation is 0.746 with both cardinality and answer frequency removed.

Nor by one domain. Eleven of the 28 relations have *person* subjects, raising the worry that one domain (i.e., one subject type) drives the result. Holding out each subject domain in turn leaves $\rho \in [0.570, 0.700]$, median 0.656.

Unchanged findings under an alternative scoring protocol We reran the analysis in Figure 1 under an alternative protocol: a single shared split and objects restricted to those their first token identifies. This gives $\rho = 0.644$ ($p = 0.0005$) over the 25 relations that meet the restriction, against 0.658 over 28 here.

A within-domain check. PERSON_FATHER and PERSON_MOTHER are separate relations over near-identical subject sets. They receive almost identical faithfulness (0.09, 0.07) and almost identical lens scores (-3.69, -3.99), without the lens being told which relation it is looking at.

6 Where the account degrades

What the lens reads is what an entity is *salient* for: a relation is linearly decodable to the extent that its answer is already part of what the subject's representation makes the model disposed to say. The relations that deviate from the trend are therefore exceptions to the account. Here, we ask whether these relations are unstructured.

Salience, measured independently of the lens. our account would be circular if salience were read off the lens, since the lens is itself a salience readout. We therefore measure it by the model's own free generation. For each test subject we greedily generate 90 tokens from each of `"{} is best known for"` and `"Here are some facts about {}. First,"`, and record whether any distinctive word of the gold object appears in either continuation.[2] This is a forward pass under different prompts, and involves no Jacobian.

We refer to this rate as the object's *prominence*. Across the 14 relations for which it can be measured it ranges from 1.00 (CITY_IN_COUNTRY) and 0.50 (COUNTRY_CAPITAL_CITY) down to 0.13 (PERSON_FATHER), 0.05 (SUPERHERO_ARCHNEMESIS) and 0.03 (PERSON_MOTHER). It predicts how often the lens surfaces the gold object in its top five: Spearman $\rho = 0.614$ ($p = 0.019$, $n = 14$). Table 1 gives both quantities per relation. The salience interpretation is thus supported by a measurement that is independent of the quantity it explains.

One relation is an outlier: STAR_CONSTELLATION has prominence 0.60 yet the lens surfaces the constellation for only 5% of

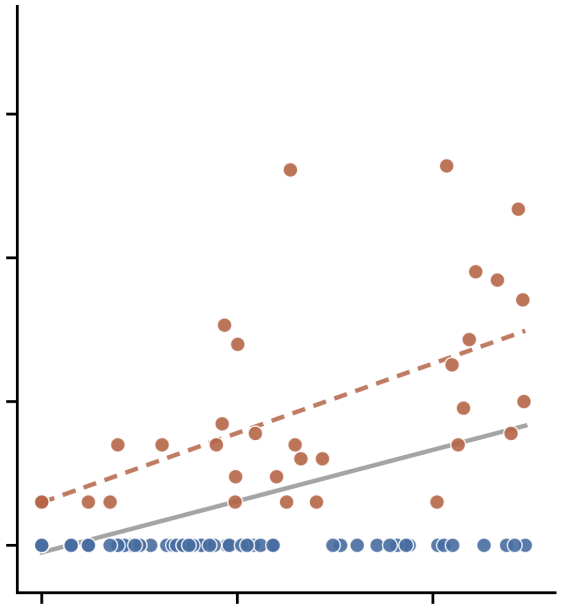


Figure 2: Rank-rank analysis of individual factors for PRODUCT_COMPANY. x-axis: lens $\log_1 0$ rank. y-axis: LRE $\log_1 0$ rank. Faithfulness: 0.68. Grey regression line over all points (n=99, $\rho = .41$). Orange regression line over orange points (misses: n=32, $\rho = .58$).

[1] All variables here are ordinal, so we work throughout with ranks. To remove the effects of a set of covariates, we rank-transform the lens score, faithfulness, and each covariate; regress the ranked lens score and ranked faithfulness separately on the ranked covariates using ordinary least squares with an intercept; and report the Pearson correlation between the resulting residual vectors. This is the standard partial Spearman correlation, and it asks how much of the lens–faithfulness association remains once everything the covariates can linearly explain about either variable has been removed.

[2] Titles, particles and words shared with the subject's own name are excluded, so that *The* Joker is not matched by any occurrence of "the" and *Johann Georg Kant* is not matched by a passage about Immanuel Kant. The lens's own prompt cannot be used here: `"{} is known for"` makes this model emit a multiple-choice quiz rather than prose, so the object is never named.

relation	prom.	n	gold∈lens	n
city_in_country	1.00	5	0.65	17
person_plays_pro_sport	0.60	40	0.35	20
star_constellation*	0.60	40	0.05	20
country_capital_city	0.50	8	0.56	16
landmark_in_country	0.47	40	0.25	20
product_by_company	0.33	40	0.05	20
person_occupation	0.30	40	0.10	20
country_currency	0.29	14	0.00	20
person_university	0.28	29	0.00	20
country_largest_city	0.25	8	0.25	16
person_father	0.13	39	0.06	18
company_hq	0.12	40	0.00	20
superhero_archnemesis	0.05	39	0.05	19
person_mother	0.03	40	0.00	20

Table 1: Object prominence, measured by the model's own free generation and therefore independently of the lens, against how often the lens surfaces the gold object in its top five. Spearman $\rho = 0.614$ ($p = 0.019$, $n = 14$). *: star_constellation is an outlier.

its stars. We suspected an artifact, since a Bayer designation contains the constellation in the Latin genitive (*Eta Persei* for *Perseus*), which our word filter does not remove; but prominence on the 19 subjects without such overlap is 0.63, slightly higher rather than lower, so this explanation fails. What the lens returns for these subjects is category vocabulary – *constellation*, *brightness*, *stellar* – rather than the particular constellation. Excluding the relation would raise the correlation to 0.725.

They are not. Ordering relations by their residual from the fitted trend, both tails are organized by the size of the answer space. The largest positive residuals – faithful despite a low lens score – are LANDMARK_ON_CONTINENT ($+0.30$, $k = 189$), NAME_GENDER ($+0.29$, $k = 9.5$) and PERSON_NATIVE_LANGUAGE ($+0.26$, $k = 30.6$). In each, the LRE chooses among a handful of possible answers, and can be faithful without the answer being salient for any particular subject: no first name is chiefly "known for" its gender, but there are only two answers to pick between. The largest negative residuals – unfaithful despite the lens – are PERSON_FATHER (-0.43), PERSON_MOTHER (-0.41) and COMPANY_CEO (-0.32), all with $k \approx 1$ and answer spaces in the hundreds, where faithfulness is near its floor whatever the representation carries.

The account thus holds in the middle of the cardinality range and degrades predictably at both ends, in opposite directions. This is the same variable whose removal raises the correlation to 0.744 (§5), and it gives that number an interpretation: k is the principal source of deviation in the salience account rather than merely a nuisance correlate.

7 Fact-level analysis

Linearity emerges as a phenomenon at the relation level, not the fact-level – it would not make sense to investigate linearity for the facts of a relation that is not represented linearly. But once there is evidence for at least partial linearity of a relation, we can ask for individual facts how well the differential Jacobian lens predicts the degree to which they are integrated into the linear structure.

We use the setup shown in Figure 2 to investigate fact-level linearity. The figure shows that lens rank is a moderately good predictor of LRE rank for PRODUCT_COMPANY ($\rho = .41$), especially if we only consider misses ($\rho = .58$). Points with an LRE rank of 0 (i.e., a hit, not a miss) are "censored": They give us no resolution because they fail to distinguish cases where the LRE predicts the correct object perfectly vs those where it barely avoids a miss. So misses may be a better diagnostic for assessing the size of the correlation.

Our original plan was to develop a fine-grained account of how differential Jacobian lens readouts influence the linearity of individual facts. However, we found that several shortcomings of the experimental setup currently used for investigating relations in LMs make this hard. In this section, we analyze these shortcomings and give guidance for future relation datasets that may better support research on how LMs represent facts.

We conduct our analysis on the three relations that have the worst overall correlation in the rank-rank analysis: their correlations are close to zero or even negative. See Figure 3.

7.1 Semantically weak first tokens

PERSON_UNIVERSITY exemplifies the problem of *semantically weak first tokens*. Many universities have names of the form "University of". So the LM learns to predict "University" as the target token for many persons. The standard setup is to count such a prediction as correct. But this is not an effective evaluation: The correct generation of "University" is not a reliable indicator of the factual competence of the LM.

Measuring the collapse rate – the fraction of a relation's distinct objects that share a first token with another – **6 of 40 relations exceed 25%**, led by PERSON_MOTHER (61%), PERSON_FATHER

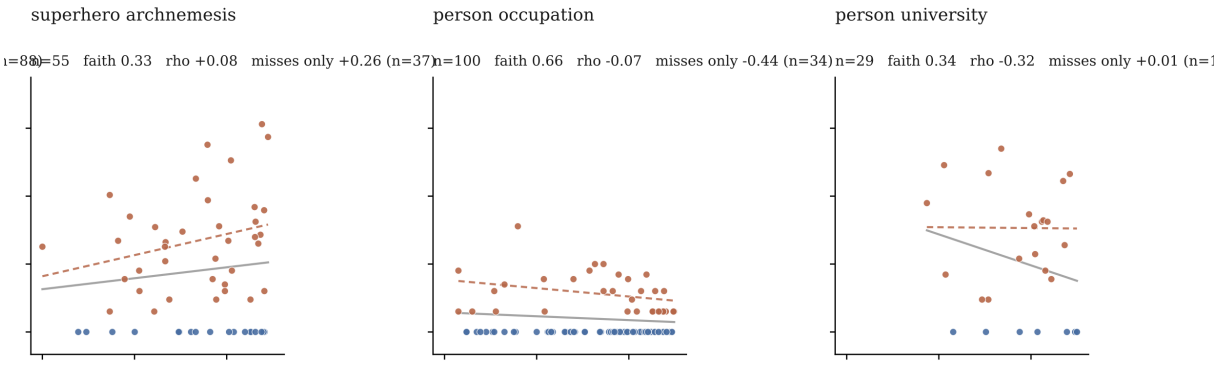


Figure 3: Rank-rank analysis for the three relations with the worst overall corelation between lens rank and LRE rank. Grey regression lines over all points, orange regression lines over orange points.

(58%) and PERSON_UNIVERSITY (45%, where 29 of 69 objects begin with UNIVERSITY).

Restricting each relation to objects whose first token identifies them repairs this on the gold side, and is the scoring used in §7. It is only a partial remedy, because faithfulness never uses gold: it compares the LRE against the *model's* top-1, so what inflates it is the model's own answer being an easy token. In PERSON_UNIVERSITY, after filtering, the model still answers UNIVERSITY for 7 of 29 subjects and the LRE agrees on every one. A complete fix would condition on the prediction being distinctive.

The direction of the bias differs by relation. Where the collapse is on the gold side it makes faithfulness easier – 41 distinct fathers all counting as JOHN – so our PERSON_FATHER figure is an upper bound, and the corrected value is lower still ($0.07 \rightarrow 0.05$). Where the model itself emits the shared token, the collapse manufactures agreement that is not retrieval at all.

We flag all of this because the failure is silent: it produces a perfect or inflated score rather than an obvious error, and it will affect any first-token evaluation whose answers share prefixes – numeric, titled (*Doctor Doom*, *The Joker*) or generic (*University of X*).

Both LRE and the lens work on the token-level, the residual stream vector at the last layer, which is focused on producing the correct next token and may not represent reliable information about the tokens to follow. This single-token regime is highly problematic for the standard setup because we are effectively investigating a relation in which all objects have been shortened to their first token. This might be a nonsensical relation even to humans and it is clearly different from the intended relations whose objects are full entity names.

8 Subject ambiguity

PERSON_OCCUPATION exemplifies the problem of *subject ambiguity*. An example is Rolf Thommessen who is listed as an actor in LRE. The highest ranked token given by the lens is "whistlebl". This indicates that the most prominent entity associated with the last name Thommessen is Advokatfirmaet Thommessen, a law firm known for their digital whistleblowing tool. One might hope that the first name "Rolf" disambiguates which subject is intended, but apparently this is not the case. In general, the last few tokens seem to dominate which entity is represented and used by the LM. For example, for the actor Barry Sobel the lens top-ranks "kosher", "Jewish", "Rabbi", correct associations for the Brazilian Rabbi Henry Sobel; for the actress Vondell Darr, the lens top-ranks "criminal", "murders", "felon", plausible associations for the wrongfully convicted Acacia Lyn-Darr; for the actor Simon Gosejohann, the lens top-ranks "Johann", "Hann", "German", suggesting the LM simply uses the string "Johann" as the representation. While we did not conduct an extensive analysis, it is likely that subject ambiguity is a problem for many relations whose subjects are persons. Many of those persons share the properties that result in Barry Sobel, Vondell Darr and Simon Gosejohann being misrepresented by the model.

In summary, these three relations have these problems and these problems seem to occur throughout the dataset. WHAT NEEDS TO BE DONE.

9 Discussion (just random stuff for now)

The instrument we use to show this is new, and we offer it as a contribution in its own right. A *Jacobian lens* propagates a first-order approximation of the model's own computation from an intermediate layer to the output vocabulary; recent work uses one to characterize what a model is internally disposed to say, averaging a Jacobian over a corpus and applying it to raw activations (Gurnee et al., 2026). That is the right construction for describing a model's dispositions in general. It is the wrong one for describing a single entity, because a raw activation is dominated by what every entity of its type shares – the type itself, the syntactic role, the prompt – so the entity is buried in its own category.

Why differences. The quantity we want is not what the model says about France, but what is *specific* to France among countries. A raw activation s_e is dominated by what all entities of its type share – the type itself, the syntactic role, the prompt – and a lens applied to it returns mostly that shared content.

We note that this differencing falls directly out of the rewriting of the Jacobian lens in Eq. 3, i.e., out of expressing the LRE as a constant plus a sum of $J_i(s - s_i)$ terms. Eq. 4 is that same quantity with the subject-independent anchor dropped and $\beta = 1$. This new way of looking at the Jacobian lens allows us to estimate the lens without any relation and ask whether it predicts an LRE that was fitted *with* one. This also distinguishes our lens from that of Gurnee et al. (2026), which averages a Jacobian over a corpus and applies it to raw activations.

10 Limitations

The study covers one model and 28 relations. The relationship is correlational: we show that a relation-agnostic readout predicts LRE faithfulness, not that the representation it reads is what the LRE exploits.

The fact-level analysis of §7 carries several caveats of its own. Model confidence predicts LRE success better than the lens does ($11.5\times$ against $6.7\times$), and explains more of the within-relation variance (0.075 against 0.047). The lens contributes independently of it, but neither explains much: together they account for under a tenth of the within-relation variation in LRE success. Faithfulness there is estimated from a single split per relation, and a single split is noisy – one relation scored 0.39 and 0.21 on two draws at identical test-set size – so the outcome variable is measured with error that attenuates the association. The thirteen qualifying relations all sit in the middle of the cardinality range, where the account is expected to work, making this a fair test of the mechanism but not of its boundaries. And the inclusion criteria screen on faithfulness and test-set size, neither of which catches a relation like PERSON_OCCUPATION where two answers cover 65% of subjects; screening on the concentration of the model's answer distribution would be better, and would have to be fixed in advance.

The account has a scope condition we can state but not yet explain (§6). It is weakest where the answer space is very small, since an LRE can then be faithful by elimination rather than by reading anything from the subject, and where the answer space is very large and each answer near-unique, since faithfulness is then close to its floor regardless. We show that these are the two tails of a single variable, but we do not have a principled account of where the boundaries lie, and with 28 relations we cannot fit one.

We also probe salience through a single neutral prompt, `"{} is known for"`. That phrasing asks what an entity is *noted* for, which need not coincide with everything its representation carries; attributes such as gender may be well encoded without being something a name is "known for". A different neutral prompt, or an average over several, might track salience more broadly.

Seven of the 28 relations have fewer than ten test items after the in-context and reference subjects are removed, so their faithfulness estimates are coarse (for three of them, quantized to thirds); a rotating-split protocol would sharpen these. Finally, the diagnostic is defined only for relations whose subjects are entities, which excludes 19 of the dataset's 47 relations.

11 Conclusion

Whether a relation is linearly decodable from a language model is substantially predictable without the relation. A Jacobian lens estimated from subject differences under a neutral prompt, at a fixed layer, ranks the model's own answer highly for exactly those relations whose LREs turn out to be faithful.

The reading we favor is that salience is the mechanism rather than a confound: a relation admits a

faithful linear operator to the extent that its answer is already part of what the subject's representation makes the model disposed to say. On that view the wide spread of LRE faithfulness across relations is less a fact about the relations than about the entities they are asked about – and the exceptions are informative, falling at the two ends of the answer-space range where faithfulness can be achieved, or prevented, without the subject's representation mattering much either way.

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A Jacobian–vector products

Eq. 4 needs $J_i u$, not J_i. We obtain it by the standard double-backward construction (Pearlmutter, 1994): differentiate $\langle u', f(h)\rangle$ with respect to h with u' held as a free variable, then differentiate the result with respect to u' in the direction u. Each product therefore costs a constant number of forward and backward passes rather than the d_{model} passes a materialized Jacobian would need. The construction requires `float32` and eager attention, as the fused attention kernels are not twice differentiable.

B Layer selection

Layer 17 is roughly half-way through the network, the region in which such lenses are usable at all (Gurnee et al., 2026), and is the modal best layer for the LRE itself in our sweep (10 of 28 relations; the rest spread over 5, 11, 14, 20 and 23). We stress that a single layer is used for *every* relation, so this choice cannot manufacture a correlation across relations: it can only make the diagnostic uniformly better or uniformly worse. Tuning the layer per relation would score higher and mean less.

C Permutation test for the main correlation

The correlation in §5 rests on 28 points, and its parametric p-value assumes more than we would like. We therefore also test it by permutation: we shuffle the lens scores against the faithfulness values 10^4 times, recompute ρ each time, and count how often the shuffled $|\rho|$ matches or exceeds the observed one. It does so in 0.03% of shuffles, so the association is not an artifact of the small sample.

D All relations

Table 2 gives faithfulness and the lens score for every relation in the study. Table 1 gives object prominence and the lens's top-five hit rate for the 14 relations on which prominence can be measured (Section 6).

relation	k	faith.	A_r	relation	k	faith.	A_r
country_language*	1.7	1.00	-1.04	presidents_election_year†	1.1	0.67	-4.17
country_largest_city*	1.0	1.00	-0.81	product_by_company	17.4	0.66	-3.24
landmark_on_continent	189.4	0.94	-2.81	person_plays_position_in_sport	68.0	0.64	-3.99
person_native_language	30.6	0.94	-2.56	person_band_lead_singer*	1.0	0.60	-1.68
name_birthplace	3.9	0.93	-1.29	person_university	1.3	0.59	-4.52
name_religion	6.2	0.93	-1.37	company_hq	4.1	0.52	-4.09
person_plays_pro_sport	63.6	0.92	-1.45	superhero_person	1.0	0.48	-2.15
person_plays_instrument	85.5	0.88	-1.77	star_constellation	11.7	0.42	-3.67
landmark_in_country	9.2	0.84	-2.25	pokemon_evolutions	1.0	0.39	-3.44
city_in_country	1.3	0.82	-0.48	country_currency	1.3	0.36	-4.48
person_occupation	26.5	0.76	-2.81	superhero_archnemesis	1.1	0.25	-3.57
country_capital_city*	1.0	0.75	-0.74	person_father	1.0	0.09	-3.69
name_gender*	9.5	0.67	-4.67	company_ceo	1.0	0.07	-4.59
presidents_birth_year†	1.3	0.67	-4.24	person_mother	1.0	0.07	-3.99

Table 2: All 28 relations, sorted by LRE faithfulness. k is the mean number of subjects per distinct object; A_r is the relation-agnostic lens score, higher meaning closer agreement with the model. †rescored for the numeric-object artifact (Appendix ?); *fewer than ten test items, so faithfulness is coarsely quantized.